



Delivering individual, specific
ultra high-speed solutions that
meet our clients needs exactly,
whatever the application.

That's our commitment.



Our company

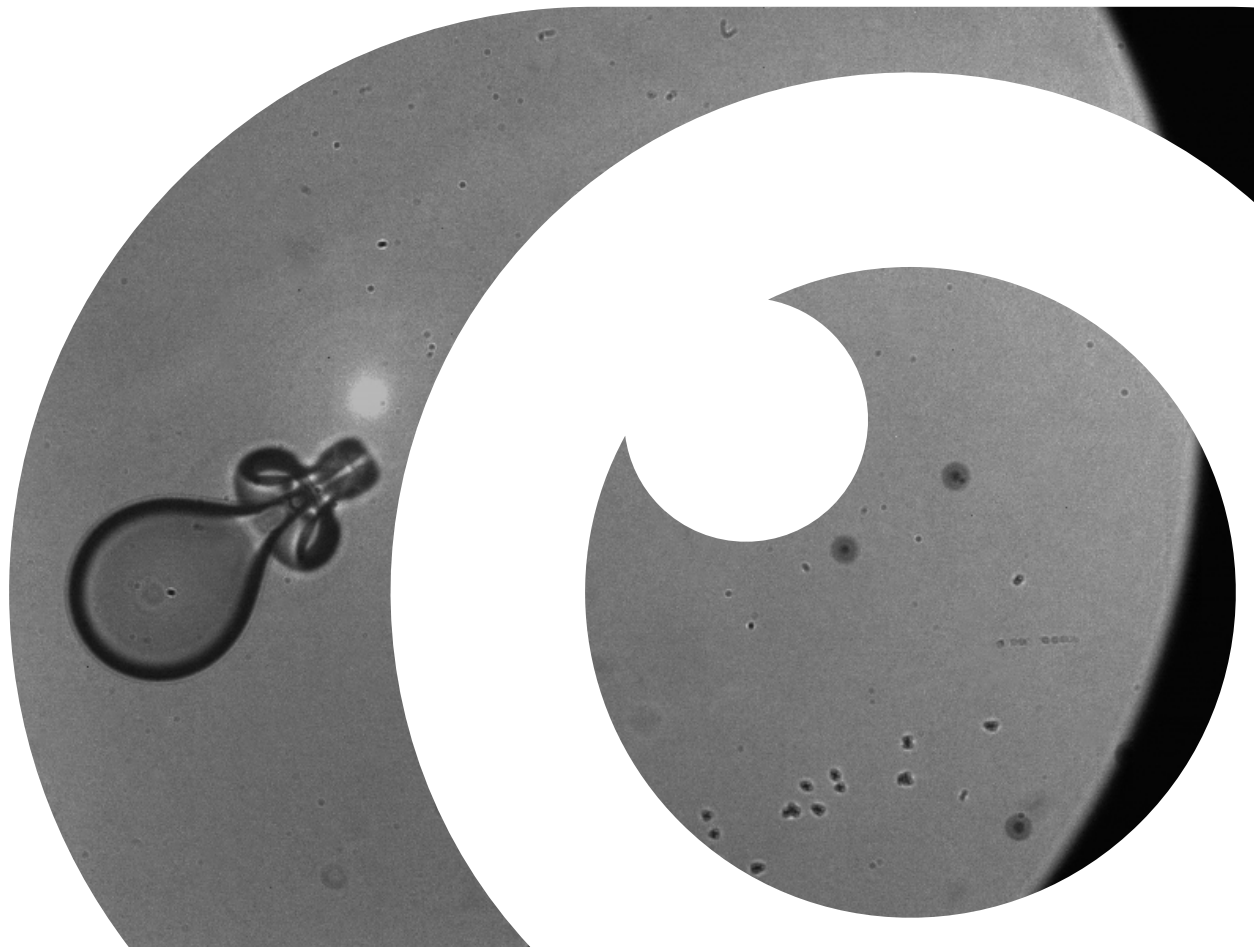
Specialised Imaging is an internationally renowned company that focuses on the design and manufacture of ultra-high-speed imaging cameras for industrial, scientific and defence research applications.

The company was formed in 2003, its founder members having previously worked together in the high-speed imaging field and bringing over 80 years' combined experience to the venture.

Specialised Imaging has successfully launched many new and innovative ultra-high-speed imaging systems.

The company is at the forefront of world-wide innovation in the high-speed imaging field, having won the BEEA's Small Company of the Year award in 2009, the Queen's Award for Enterprise in 2011 and 2016.

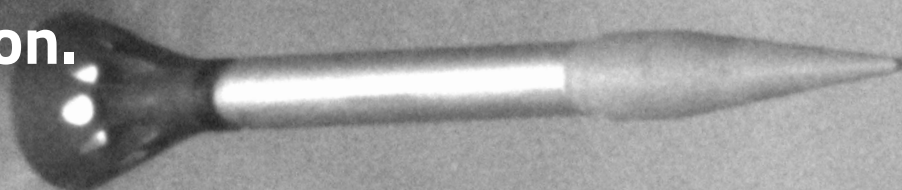
This commitment to development has enabled the company to establish a reputation as an exciting and creative player in the high-speed camera market.





Innovative imaging solutions
that incorporate the latest
technological advances.

That's our passion.



The Ultimate High-Speed Video Camera



Up to 20 Million images/second capture speed

900 x 740 pixel, 10-bit sensor resolution

180/360 images

25ns minimum exposure

The Specialised Imaging AURORA is the latest Ultra high-speed video camera that combines the flexibility of a video camera with the speed/resolutions approaching those only available with Framing cameras.

The unique custom design sensor offers 180 images at capture speeds up to 20 Million Images/second at full resolution. Now offers 360 images at capture speeds up to 20 Million Images/second at 900x370 pixel resolution.

In line with high-speed video cameras the AURORA can be Frame synchronised with an external device such as another KIRANA, AURORA or laser. The AURORA can also be recording prior to the event and triggered before, during or after the event.

FEATURES

- ☐ Up to 20 Million images/second
- ☐ Adjustable exposure down to 25ns
- ☐ Novel, patented sensor architecture
- ☐ Pre & Post event triggering
- ☐ External synchronisation
- ☐ Nikon lens mount fitting
- ☐ Gigabit ethernet communications
- ☐ Compact and rugged design

Aurora5M Up to 5 Mfps

Aurora10M Up to 10 Mfps

Aurora20M Up to 20 Mfps

MODEL SPECIFICATION

	Aurora5M	Aurora10M	Aurora20M
Frame Rate (Frames per second)	Up to 5 Mfps	Up to 10 Mfps	Up to 20 Mfps
Exposure Time (minimum)	25ns		
Trigger Mode	Start on Trigger, Stop on Trigger (user defined post trigger frames)		

OPTICAL

Lenses	Nikon F-Mount
Shutter	Electro-mechanical
Distortion	Zero

INTENSIFIER / SENSOR

Sensor	μCMOS - BSI (Patented Technology)	
Number of Active Pixels	900 (W) x 740 (H)	900 (W) x 370 (H)
Number of Frames	180 Frames	360 Frames
Pixel Size	30μm	
Digitisation	10bits	

MECHANICAL

Dimension mm (w/d/h)	Head: 22.8cm x 42cm x 19cm (without lens) Power supply: 19.5cm x 39.5cm x 19.5cm (inc. handle)
Weights	Head: 10.6Kg (23lbs) without lens. Power Supply: 4.8Kg (10.5lbs)
Tripod Mounting	3/8-16 UNC Female in base.

ENVIRONMENTAL

Storage temperature	-10°C to +50°C
Operating temperature	-5°C to +40°C
Humidity	10—90% RH non condensing
Vibration shock	10—40 Hz Max. 10g in any direction
EMC	Meets all UKCA/EU harmonised standards

TIMING PARAMETERS

System Clock	200MHz quartz crystal controlled
Exposure time	25ns
Framing rates	1000fps - 5Mfps / 10Mfps / 20Mfps

INPUT / OUTPUT SIGNALS

Trigger (2 off) (BNC)	Electrical signal Threshold variable from ± 25V Maximum Input level 50V Integrated velocity timing system Positive or Negative polarity, Make/Break 50Ω or 1KΩ termination
Video Out	XVGA
Aux Out (x4) (BNC)	FSync or user programmable pulse width and position for strobe/laser illumination sources. TTL into 50Ω
Sync In	Input to allow the synchronisation of two cameras in Master-Slave configuration
Camera Control	Remote control via Standard 1Gbps Ethernet
Software	Custom software compatible with Microsoft Windows Operating Systems for control and data archiving in various file formats
Electrical input	Mains 100-240V AC 50-60Hz
Saved Image Format	TIFF, JPEG, AVI or RAW



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Up to 1 Billion frames per second capture speed



The Specialised Imaging SIM Framing Camera range offers from 16 to 32 images without creating shading, or parallax. Highly accurate timing and fully flexible intensified CCD sensors provide almost infinite control over interframe time, gain and exposure to capture even the most difficult ultra-fast phenomena.

Comprehensive triggering adjustment and a wide range of output signals are controlled using the custom software package which also includes measurement and image enhancement functions.

The SIM has an optional port for the addition of a high-speed video, or streak camera to allow simultaneous long duration or ultra high temporal resolution capture.

FEATURES BOTH MODELS

- ☐ 1360 x 1024 pixel sensor resolution
- ☐ Up to 16 discrete intensified optical channels
- ☐ Adjustable inter-frame time in 1ns steps
- ☐ Fully adjustable exposure down to 3ns
- ☐ Gain adjustment up to 10,000X
- ☐ Programmable output triggers
- ☐ Nikon lens mount (standard)
- ☐ Canon lens mount (optional)
- ☐ Gigabit Ethernet communications
- ☐ High system resolution configuration
- ☐ User interchangeable filters

HIGHER RESOLUTION

SIMX**SIMX4 SIMX8 SIMX10 SIMX12 SIMX16**

Number of Channels	4	8	10	12	16
Number of images	4	8	10	12	16

Single and multiple channel upgrades are available up to a maximum of 16 channels.

MORE IMAGES

SIMD**SIMD8 SIMD16 SIMD20 SIMD24 SIMD32**

Number of Channels	4	8	10	12	16
Number of images	8	16	20	24	32

Single and multiple channel upgrades are available up to a maximum of 16 channels.

SPECIFICATION BOTH MODELS

OPTICAL

Optics	Single input beam splitting optics
Lenses	Nikon F-Mount (Standard) Canon Mount (Optional)
Internal electro-mechanical iris	f2.8 - f22
Shutter	Electro-mechanical
Distortion	Nominally zero
Channel Registration	Within one pixel after software correction
Intensity Variation	Better than 5% across the image

INTENSIFIER / SENSOR

Image Sensor	ICX285AL
Active CCD Pixel	1360 (H) x 1024 (V)
Pixel Size	6.45 µm (H) x 6.45 µm (V)
Digitisation	12 bits
Intensifier	Gen II 18mm High resolution MCP Input window Fused Silica Output window Fibre Optic Photocathode S25, others available on request SIMD Phosphor screen P46 SIMX Phosphor screen P43

Other Image Intensifier options available. Please contact us to discuss.

Gain	Variable up to 10,000
System resolution	>50 lp/mm (SIMX) >36 lp/mm (SIMD)

MECHANICAL

Dimensions in cm (LxWxH)	57.2 x 43.8 x 31.9 (> 8CH, without lens) 57.2 x 23.8 x 31.9 (< 8CH, without lens)
Mount	3/8-16 UNC Female
Weight	37.5Kg (< 8CH, without lens) 24Kg (> 8CH, without lens)

TIMING PARAMETERS

System Clock	1GHz quartz crystal controlled
Exposure Mode (each image)	Single exposure or multiple exposures (Max. 8) per channel
Exposure Time	3ns - 10ms in 1ns steps independently variable
Separation Time (multiple exposure mode)	30ns - 20ms in 1ns steps independently variable
Interframe Time	0ns - 20ms in 1ns steps independently variable
Delay to 1st exposure	65ns to 10ms in 1ns steps, independently variable
Flash Outputs	5ns - 1ms in 1ns steps independently variable
Framing rates	up to 1 Billion fps

INPUT / OUTPUT SIGNALS

Trigger 1 (BNC)	Electrical signal Threshold variable from $\pm 25V$ Positive or Negative polarity, Make/Break 50Ω or 1KΩ termination
Trigger 2 (BNC)	Electrical signal Threshold variable from $\pm 25V$ Positive or Negative polarity, Make/Break 50Ω or 1KΩ termination
Timing Monitor Pulses (BNC)	Pulse width (min. 3ns) and position user programmable TTL into 50Ω
Flash Trigger Outputs	Pulse width (min. 5ns) and position user programmable TTL into 50Ω
Camera control	Data and command transfer via Gigabit ethernet cable length 10m (standard), other lengths up to 100m. Optional Fibre Optic ethernet link (up to 2Km)
Software	Custom software compatible with Microsoft Windows Operating Systems for camera control, image data archiving in various file formats.
Electrical input	Mains 100-240V AC 50-60Hz

ENVIRONMENTAL

Storage temperature	-10°C to +50°C
Operating temperature	-5°C to +40°C
Humidity	10 - 90% RH non condensing
Vibration shock	10 - 40 Hz Max. 10g in any direction
EMC	Meets all UKCA/EU harmonised standards

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High Resolution Double Image Intensified camera



Single or Double image capture

The Specialised Imaging SIR3 Framing Camera offers up to 2 high resolution images, 100µs apart. Fully flexible intensified CCD sensor provides control over interframe time, gain and exposure.

Comprehensive triggering adjustment and a wide range of output signals are controlled using the custom software package which also includes measurement and image enhancement functions.

Up to 10.7 MegaPixel 12-bit images

Lightweight and Rugged construction

FEATURES

- ☐ 100µs minimum delay between 2 images
- ☐ Fully adjustable exposure down to 10ns
- ☐ Gain adjustment up to 10,000X
- ☐ Adjustable output triggers
- ☐ Nikon lens mount fitting
- ☐ Gigabit ethernet communications

OPTICAL

Lenses	Nikon F-mount (ruggedized mounting system)
System Aperture	f 2
Shutter	Electro-mechanical
Distortion	Nominally zero
Coupling	Optical
Vignetting	<3%
Intensity variation	Better than 5% across the image
Optical Viewfinder	Optional

INTENSIFIER / SENSOR

	SIR3-25D	SIR3-40D
Image Sensor	KAI4021M	KAI11002M
Active CCD Pixel	2048 (H) x 2048 (V)	4008 (H) x 2688 (V)
Pixel Size	7.4 µm (H) x 7.4 µm (V)	9 µm (H) x 9 µm (V)
Dynamic Range	12 bits	12 bits
Intensifier diameter	25mm MCP	40mm MCP
Photocathode	All models: S25	
Phosphor / decay	FS/10µs	FS/10µs
Input / Output windows	All models: Glass / Fibre	
Gain	Variable up to 10,000 all models	

MECHANICAL

Dimension mm (w/d/h)	17.0cm x 48.5cm x 19.3cm (without lens)
Tripod Mount	1/4 - 20 UNC and 3/8 - 16 UNC female
Weight	15Kg (33lbs) without lens

ENVIRONMENTAL

Storage temperature	-10°C to +50°C
Operating temperature	-5°C to +40°C
Humidity	10—90% RH non condensing
Vibration shock	10—40 Hz Max. 10g in any direction
EMC	Meets all UKCA/EU harmonised standards

TIMING PARAMETERS

System Clock	200MHz quartz crystal controlled
Inherent Delay	<130ns
Imaging Mode	Single or Double image
Exposure Modes	Single exposure Multiple exposures (Max. 16 - subject to imaging conditions).
Exposure Times	10ns – 10ms in 5ns steps independently variable
Separation (Multiple Exposure Mode)	30ns to 20ms in 5ns steps independently variable
Delay to 2nd Image	100µs – 10ms in 5ns steps.
Flash output	20ns to 1ms in 5ns steps independently variable

INPUT / OUTPUT SIGNALS

Trigger 1 (BNC)	Electrical signal Threshold variable from ± 25V Positive or Negative polarity, Make/Break 50Ω or 1KΩ termination Maximum Voltage +/- 50V
Trigger 2 (BNC)	Electrical signal Threshold variable from ± 25V Positive or Negative polarity, Make/Break 50Ω or 1KΩ termination Maximum Voltage +/- 50V
Flash Trigger Output (BNC)	Pulse width (min. 10ns) and position user programmable. TTL into 50Ω
Camera Control	Data and command transfer via Gigabit Ethernet Cable length 100m (standard) 1000FX fibre optic Ethernet link (up to 2Km) - optional
Software	Custom software compatible with Microsoft Windows Operating Systems for camera control, image data archiving in various file formats.
Electrical input	Mains 100-240V AC 50-60Hz

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SI-OT4



Optical Trigger Unit



Rugged Design

Optimisable Background, High Contrast detection

Shadow & Flash Detector

The Specialised Imaging OT4 provides a reliable optical trigger for either projectile "shadow" detection or flash detection. Our low-ripple AC power unit, with a rugged enclosure allows the OT4 to be used outside in all weathers.

FEATURES

- ☐ Small and lightweight
- ☐ Low ripple rugged AC power unit
- ☐ Nikon lens mount fitting
- ☐ User adjustable sensitivity



OPTICAL

Lenses	Nikon F-Mount
Alignment	Optical viewport

SENSOR

Sensor	Multi-Segment Photodiode array. 400nm – 700nm range
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INPUT / OUTPUT SIGNALS

Output (BNC)	Positive 5V
Trigger indicator	LED
Electrical input	18-24V DC AC Power Unit

MECHANICAL

Dimension mm (L x W x H)	OT4 Unit (without lens) 160mm x 140mm x 85mm 6.3" x 5.5" x 3.3"
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AC Power Unit 150mm x 120mm x 80mm 5.9" x 4.7" x 3.1"
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Weights	OT4 Unit 1.1Kg / 2.4lb AC Power Unit 2Kg / 4.4lb
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Sensor head mounting	1/4 - 20 UNC Female Base and top
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ENVIRONMENTAL

Storage temperature	-10°C to +60°C
Operating temperature	-5°C to +45°C
Humidity	10—90% RH non condensing
Vibration shock	10—40 Hz Max. 10g in any direction
EMC	Meets all UKCA/EU harmonised standards



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Compact image intensifier/ converter

**Range of Intensifier
tube options**

**Up to 40lp/mm
resolution**

**Gate times down
to 10ns**

**Comprehensive triggering for
High-speed video and other cameras**



The Specialised Imaging SIL3 image intensifier is the latest generation of intensifier that can be synchronised to almost any high-speed video, conventional video or stills camera.

Maximum equivalent frame rates ranging from 100,000fps to 1,000,000fps and minimum gate (exposure) time of 10ns.

All Intensifier tube options use fused silica input windows as standard. The two possible output window diameters, Phosphor decay times and Wavelength responses options (inc. UV) allows the SIL3 to be configured as a unique enhancement to any camera.

Intuitive custom control software allows gain, gate/ exposure and delay adjustment via Ethernet.

FEATURES

- ☐ 18mm, 25mm & 40mm Intensifier tube options
- ☐ Fully adjustable gating times down to 10ns
- ☐ Gain adjustment up to 500,000X
- ☐ Nikon lens mount fitting
- ☐ Optical camera coupling
- ☐ Ethernet communications
- ☐ UV wavelength conversion option

MODEL SPECIFICATION

SIZE	Gain	Minimum Gating Time	Photo-cathode	Phosphor	"Max. Frequency (Hz)"	"Typical Resolution (lp/mm)"	"Typical Resolution (μm)"
SIL3 18	NG (7,000)	50	S20UV	P46	1M	50lp/mm	10
		20	S25				
		10					
	HG (70,000)	50	s20UV	P46	1M	35lp/mm	14.3
		20	S25				
		20					
SIL3 25	NG (10,000)	50	S20UV	FS	100K	40lp/mm	12.5
		20	S25	P46	1M	32lp/mm	15.6
	HG (100,000)	50	S20UV	FS	100K	28lp/mm	17.8
		20	S25	P46	1M	22lp/mm	23
	EG (500,000)	50	S20UV	FS	100K	26lp/mm	19.2
		20	S25	P46	1M	20lp/mm	25
SIL3 40	NG (10,000)	50	S20UV	FS	100K	28lp/mm	17.8
			S25	P46	1M	22lp/mm	23
	HG (100,000)	50	S20UV	FS	100K	20lp/mm	25
			S25	P46	1M	16lp/mm	31.25
	EG (500,000)	50	S20UV	FS	100K	19lp/mm	26.3
			S25	P46	1M	15lp/mm	33

OPTICAL

Front Lens mount	F-mount
Optical Coupling	F-mount

MECHANICAL

Dimension mm (w/d/h)	17.8cm x 18.5cm x 19.7cm (7.0" x 7.3" x 7.8") without lenses
Tripod Mount	3/8-16UNC Female
Weight	4.8Kg (10.6lbs) without lenses
Lens/Camera Support	Option available

ENVIRONMENTAL

Storage temperature	-10°C to +50°C
Operating temperature	-5° to +40°C
Humidity	10 - 90% RH non condensing
Vibration shock	10 - 40Hz Max. 10g in any direction
EMC	Meets all UKCA/EU harmonised standards

TIMING PARAMETERS

System Clock	200MHz, quartz crystal controlled.
Inherent Delay (5ns jitter)	60ns
Exposure Time	50ns—DC in 5ns steps independently variable
Inter Exposure Time	50ns—25ms in 5ns steps independently variable
Number of exposures	Up to 64 in pulsed mode. Unlimited in REP (synchronous) mode

INPUT / OUTPUT SIGNALS

Triggers (BNC)	2 Independent Inputs Electrical signal Maximum Input level 50V Threshold variable from ± 25V Positive or Negative polarity, Make/Break 50Ω or 1KΩ termination
Aux Outputs (BNC)	Pulse width and position user programmable (min. 5ns) TTL into 50Ω
Control Interface	Remote control via Standard 100Mbps Ethernet
Inhibit Input (BNC)	5V TTL (user input brightness protection)
Software	Custom software compatible with Microsoft Windows Operating Systems for control
Electrical input	Mains 100-240V AC 50-60Hz

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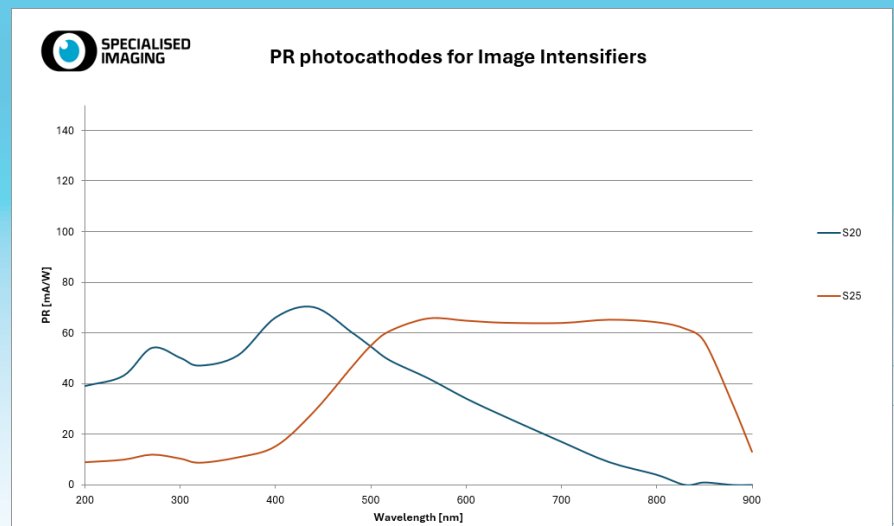
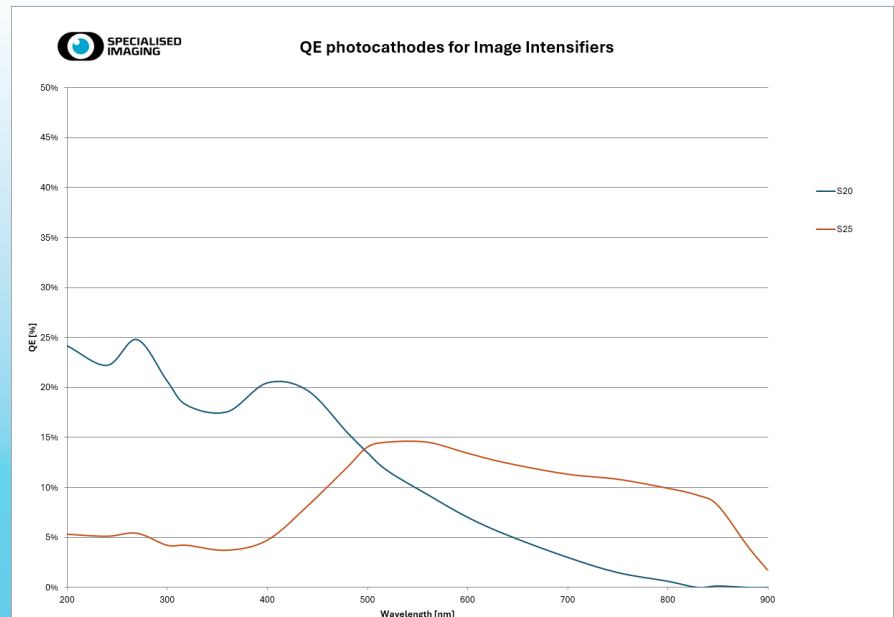
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Photocathode Spectral sensitivities:

The Fused Silica input window used in the SIL3 allows transmission down to 180nm. This feature improves the spectral response normally limited by the transmission of other input window materials, such as borosilicate or Fibre optic glass which have an inherent cut off frequency around 300 - 350nm.



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FM87429

High Resolution intensified camera



Single image capture

Up to 1.3 MegaPixel

The Specialised Imaging T-Cami offers high resolution single or double image intensified monochrome images.

Comprehensive triggering adjustment and a wide range of output signals are controlled using the custom software package which includes measurement and image enhancement functions.

12-bit images

Compact and rugged construction

FEATURES

- ☐ Fully adjustable exposure from 5ns to 10ms
- ☐ Adjustable output triggers
- ☐ Nikon lens mount fitting
- ☐ Gigabit ethernet communications

OPTICAL

Lenses	Nikon F-mount
System Aperture	Limited by lens
Distortion	Nominally zero

SENSOR

Image Sensor	ICX285AL
Active CCD Pixel	1360 (H) x 1024 (V)
Pixel Size	6.45 μ m (H) x 6.45 μ m (V)
Dynamic Range	12 bits
Intensifier diameter	18mm MCP
Photocathode	S25
Phosphor / decay	P46/300ns
Gain	Variable up to 10,000

MECHANICAL

Dimension mm (w/d/h)	17.8cm x 18.5cm x 19.7cm (7.0" x 7.3" x 7.8") without lenses
Tripod Mount	1/4-20 UNC Female
Weight	6Kg (13.2lbs) without lens.

INPUT / OUTPUT SIGNALS

Trigger 1 (BNC)	Electrical signal Threshold variable from $\pm 25V$ Positive or Negative polarity, Make/Break 50 Ω or 1K Ω termination
Trigger 2 (BNC)	Electrical signal Threshold variable from $\pm 25V$ Positive or Negative polarity, Make/Break 50 Ω or 1K Ω termination
Flash Trigger Output (BNC)	Pulse width (min. 10ns) and position user programmable. TTL into 50 Ω
Camera Control	Data and command transfer via Gigabit Ethernet. Cable length 10m (standard), other lengths up to 100m available 1000FX fibre optic Ethernet link (up to 2Km) - optional
Software	Custom software compatible with Microsoft Windows Operating Systems for camera control, image data archiving in various file formats.
Electrical input	Mains 100-240V AC 50-60Hz

ENVIRONMENTAL

Storage temperature	-10°C to +50°C
Operating temperature	-5°C to +40°C
Humidity	10—90% RH non condensing
Vibration shock	10—40 Hz Max. 10g in any direction
EMC	Meets all UKCA/EU harmonised standards

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FM87429

Compact High-Speed Linear Tracker System



Based on the Trajectory Tracker² system

Multiple tracking modes

Manual positional adjustment

**Integrated with various
high speed cameras**

The Specialised Imaging Tracker^{Lite} is a smaller, lighter system that retains the core functions of the Award winning Tracker² system.

Simplification of the Tracker² system includes manual alignment adjustment with projectile flightpath, and three inputs for real time projectile velocity variation adjustment.

Mounted on heavy, stable tripod, the fully weatherproof mirror/camera housing allows smaller high-speed video cameras with long focal length lenses to be used.

Custom software controls the Tracker and triggering set up, provides calculators for camera fields-of-view and depth of focus for different cameras, lenses and Tracker locations.

FEATURES

- ☐ Multiple operating modes allow capture of decelerating, accelerating, user defined velocity profiled projectiles
- ☐ Scan ratio (SR) range from 1 to 100
- ☐ Typical scanning accuracy of $\pm 0.2^\circ$.
- ☐ Gigabit ethernet communications
- ☐ Built in camera power, communications and trigger
- ☐ No calibration required
- ☐ Optional 32 trigger inputs

OPERATING MODES

Fixed Velocity	Single trigger using assumed initial velocity
Velocity	The scan is corrected using the measured velocity from 2 detectors.
Drag / Acceleration	The scan is corrected using the measured velocities and calculated drag from 3 detector inputs.
Pre-defined profile	Programmable Velocity vs Time curve. Triggered using single trigger. Used for non-linear projectile trajectories.
Position	The scan position is corrected by detector inputs
Advanced User Functions	Specialised Imaging is prepared to customise modes of operation to user requirements.
Skewed Geometry	Allows asymmetric scan

ENVIRONMENTAL

Storage temperature	-10°C to +74°C
Operating temperature	-5°C to +45°C
Warmup Period	Not Required
Humidity	10 - 90% RH non-condensing
Operational vibration	10G, 10-40Hz Max, any direction
EMC	Meets all UKCA/EU harmonised standards

MECHANICAL

Dimensions mm (w/d/h)	650 x 230 x 310 (without tripod)
Weight	16kg / 35lbs (without camera and lens)
Tripod Mount	3/8-16 UNC Female

MIRROR

Type	Optical flat elliptical surface silvered
Size (HxW) mm	135 x 90

OPERATING PARAMETERS

Scan Ratio (SR)	1 to 100 (defined as the ratio of projectile velocity/ stand-off distance)
Scanning range (Max.)	-50° to +50°
Scanning Distance	≥ 2 x standoff distance (shortest distance between Tracker and line of flight)
Scanning Accuracy	± 0.2°
Calibration	Not required
Projectile Velocity	SR x Standoff distance
Projectile Drag	0 to 100 m/s/m
Acceleration Angle	1° - 5° depending on scan rate (defined as the angle required to accelerate the mirror from rest to full scanning speed)

INPUT / OUTPUT SIGNALS

Detector In	BNC
Number of inputs	3 (Optional 32)
Trigger In (BNC)	Make/Break, Positive or Negative edge, Threshold variable to ± 17V 50Ω or 1KΩ termination
Camera Trigger	TTL positive pulse
Communication Interface	Data and command transfer via 1Gbps ethernet cable
Software	Custom software compatible with Microsoft Windows Operating Systems for control and data archiving in various file formats
Electrical input	Mains 100-240V AC 50-60Hz

CONTROL UNIT

System Clock	10MHz quartz crystal controlled
Trigger Jitter	<1us

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FM87429

Comprehensive High-Speed Linear Tracker System



Tracker² sets the benchmark for flight-follower systems

Multiple tracking modes

Remote control motorised adjustment

Accept various high speed cameras

The Specialised Imaging Tracker² is the next generation of projectile tracking platforms for high-speed video and measurement.

Full motorised remote control of three axis rotation and multiple inputs for real-time velocity adjustment contribute to the evolution of this benchmark system.

Built on a sturdy mount, the fully weatherproofed mirror and camera housings allow a large range of high-speed video cameras and long focal length lens options.

Custom software controls the Tracker system and provides calculators for Tracker placement, camera fields-of-view and velocities.

FEATURES

- ☐ Full remote control operation
- ☐ Multiple operating modes allow capture of decelerating, accelerating, user defined velocity profiled projectiles
- ☐ Scan ratio range from 0.1 to 100
- ☐ Scanning accuracy of $\pm 0.2^\circ$
- ☐ Gigabit ethernet communications
- ☐ Built in camera power, communications and trigger
- ☐ No calibration required

OPERATING MODES

Fixed Velocity	Single trigger using assumed initial velocity
Velocity	The scan is corrected using the measured velocity from 2 detectors.
Drag / Acceleration	The scan is corrected using the measured velocities and calculated drag from 3 detector inputs.
Pre-defined profile	Programmable Velocity vs Time curve. Triggered using single trigger. Used for non-linear projectile trajectories.
Position	The scan position is corrected by detector inputs
Advanced User Functions	Specialised Imaging is prepared to customise modes of operation to user requirements.
Skewed Geometry	Allows asymmetric scan

OPERATING PARAMETERS

Scan Ratio (SR)	0.1 to 100 (defined as the ratio of projectile velocity/ stand-off distance)
Scanning range (Max.)	-60° to +60°
Scanning Distance	>=2x standoff distance (distance from the line of flight to Tracker2)
Scanning Accuracy	±0.2°
Positional Accuracy	±0.018°
Calibration	Not required
Projectile Velocity	SR x Standoff distance
Projectile Drag	0 to 100 m/s/m
Acceleration Angle	1° - 5° depending on scan rate (defined as the angle required to accelerate the mirror from rest to full scanning speed)

ENVIRONMENTAL

Storage temperature	-10°C to +74°C
Operating temperature	-5°C to +50°C
Warmup Period	Not Required
Humidity	10 - 90% RH non-condensing
Operational vibration	10G, 10-40Hz Max, any direction
EMC	Meets all UKCA/EU harmonised standards

INPUT / OUTPUT SIGNALS

Detector In	BNC
Number of inputs	8 (Optional 32)
Trigger In (BNC)	Make/Break, Positive or Negative edge, Threshold variable to ± 17V 50Ω or 1KΩ termination
Camera Trigger	TTL positive pulse
Communication Interface	Data and command transfer via 1Gbps ethernet cable length 100m (standard). Other lengths available 1000FX fibre optic ethernet link (up to 2Km) - optional
Software	Custom software compatible with Microsoft Windows Operating Systems for control and data archiving in various file formats
Electrical input	Mains 100-240V AC 50-60Hz

MECHANICAL

Dimensions mm (w/d/h)	1340 x 670 x 590 (without tripod)
Mount	Tripod Included

MIRROR

Type	Optical flat elliptical Silicon Carbide Mirror
Size (HxW) mm	135 x 90 x 2

CONTROL UNIT

System Clock	10MHz quartz crystal controlled
Trigger Jitter	<1us

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FM87429

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Single Channel Control Unit



200MHz, System Clock

5ns Resolution

The Specialised Imaging Single Channel Control Unit is built within a compact enclosure and utilises a 200MHz quartz system clock providing 5ns timing accuracy.

Accepts a wide range of input signals. The TTL output can be used to trigger a range of devices including strobes, laser and flash unit. The enable input signal can be used as a gate to allow the output pulses only when it is needed.

Enable function

1 input channel / 1 output channel

FEATURES

- ☐ Ethernet communication
- ☐ Programmable

TIMING PARAMETERS

System Clock	200MHz, quartz crystal controlled
Inherent Delay	50ns
Range	5ns to 1s in 5ns steps independently variable
Jitter (trigger)	5ns

INPUT / OUTPUT SIGNALS

Input (BNC)	Threshold variable from $\pm 25V$ Positive or Negative polarity, Make/Break 50 Ω or 1K Ω termination
Output (BNC)	Pulse width (min. 5ns) User programmable TTL into 50 Ω
Enable (BNC)	+ve TTL Enable 0V Disable
Control interface	100Mbps Ethernet
Programmable Output	Disabled Max 30 μs User selectable
Software	Custom software compatible with Microsoft Windows Operating Systems for Control and data archiving.
Electrical input	Mains 100-240V AC 50-60Hz

MECHANICAL

Dimension mm (w/d/h)	220mm x 120mm x 90mm (+15mm)
Dimension inches (w/d/h)	8.7" x 4.7" x 3.5"
Weight	2Kg (4.4lbs)

ENVIRONMENTAL

Storage temperature	-10°C to +50°C
Operating Temperature	-5°C to +40°C
Humidity	10 - 90% RH non condensing
Vibration shock	10 - 40 Hz Max. 10g in any direction
EMC	Meets all UKCA/EU harmonised standards

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FM87429

4 Channel Delay Generator



200MHz, System Clock

5ns Resolution

3 independent input channels

4 independent output channels

The Specialised Imaging 4 Channel Delay generator is built within a compact IP65 rated enclosure and utilises a 200MHz quartz system clock providing 5ns timing accuracy.

Three independent inputs accept a wide range of electrical signals, while four independent outputs provide precise triggering of peripheral devices such as framing cameras, ultra-high speed video cameras, and flash, laser, or strobe illumination systems.

FEATURES

- ☐ Ethernet communication
- ☐ 19" Rack mount option
- ☐ Software driven

TIMING PARAMETERS

System Clock	200MHz, quartz crystal controlled
Inherent Delay	50ns
Range	5ns to 1s in 5ns steps independently variable
Jitter (trigger)	5ns
Jitter (channel to channel)	Less than 250ps

INPUT / OUTPUT SIGNALS

Input/trigger channels (BNC)	3 (independent) Electrical signal (Max +/- 50V) Threshold variable from $\pm 25V$ Positive or Negative polarity, Make/Break 50 Ω or 1K Ω termination
Output channels (BNC)	4 (independent) Pulse width (min. 5ns) and position user programmable TTL into 50 Ω
Control interface	100Mbps Ethernet
Software	Custom software compatible with Microsoft Windows Operating Systems for Control and data archiving.
Electrical input	Mains 100-240V AC 50-60Hz

MECHANICAL

Dimension mm (w/d/h)	360mm x 200mm x 77mm (with handles)
Dimension inches (w/d/h)	14.17" x 7.87" x 3.03" (with handles)
IP rating	IP65
Weight	4Kg (8.8lbs)

ENVIRONMENTAL

Storage temperature	-10°C to +50°C
Operating Temperature	-5°C to +40°C
Humidity	10 - 90% RH non condensing
Vibration shock	10 - 40 Hz Max. 10g in any direction
EMC	Meets all UKCA/EU harmonised standards



Rack mount option

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FM87429

AD-500



Single or multiple head high intensity flash system



The Specialised Imaging AD500 Flash system offers the flexibility of up to four controllable high intensity flash heads for use in scientific and industrial environments.

**500J Flash head with
2ms duration**

**Up to four independent
flash heads**

40s recycle time

**Trigger options -
short or positive
pulse**

CONTROL UNIT

Signal Input	Four independent Channels
Trigger Mode	1. Independent - All channels trigger separately 2. Simultaneous - Any channel triggers all channels
Trigger Source	Short Circuit 5-100V positive pulse
Input Impedance	50Ω
Mains Input	IEC socket
Electrical input	Mains 100-240V AC 50-60Hz
Dimensions mm	(LxWxH) 220mm x 110mm x 128mm
Weight	3.7kg
EMC	Meets all UKCA/EU harmonised standards

FLASH HEAD

Light Duration (Typ)	2ms measured to 50% of peak output
Stored Charge (max)	500J
Charge Voltage	340V
Light Source	U-Shape Xenon flashtube
Rise Time	50µs
Delay (typ)	30µs
Recycling Time (typ)	40 seconds
Dimensions	(LxDia) 270mm x 170mm
Weight (Kg)	5.25kg
Lamp cable length	2.5m
EMC	Meets all UKCA/EU harmonised standards

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As part of our on-going commitment to improvement we reserve the right to alter specifications, designs or figures, without prior notice. All dimensions and weights are approximate.

SIL-9020-01-Q04

MSFH-370



High intensity flash
with dedicated fibre
optic output



5mm diameter FO output

370J Flash head with 750µs duration

Up to four independent flash heads

40s recycle time

Trigger options - short or positive edge

The Specialised Imaging MSFH370 Flash system offers the flexibility of up to four controllable high intensity flash heads for use in scientific and industrial environments.

CONTROL UNIT

Signal Input	Four independent Channels
Trigger Mode	1. Independent - All channels trigger separately 2. Simultaneous - Any channel triggers all channels
Trigger Source	Short Circuit 5-100V positive edge
Input Impedance	50Ω
Mains Input	IEC socket
Electrical input	Mains 100-240V AC 50-60Hz
Dimensions mm	(LxWxH) 220 x 120 x 90 mm
Weight	4.5 kg
EMC	Meets all UKCA/EU harmonised standards

FLASH HEAD

Light Duration (Typ)	750µs measured to 50% of peak output
Stored Charge (max)	370J
Charge Voltage	340V
Light Source	Linear spark source flashtube
Rise Time	50µs
Delay (typ)	30µs
Recycling Time (typ)	40 seconds
Dimensions	(LxDia) 270 x 170 mm
Weight (Kg)	6 kg (Including handle, legs, FO mounting)
Lamp cable length	2.5m
EMC	Meets all UKCA/EU harmonised standards

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Acoustic trigger Unit



**Built in NiMH
battery**

Compact size

**User selectable
sensitivity settings**

TTL trigger output

The Specialised Imaging SI-AT1 provides a reliable acoustic trigger for blast or shockwave detection.

Built in battery power within a compact, rugged enclosure allows the AT1 to be used outside in all weathers and independent of mains power for up to 8 hours.

FEATURES

- ☐ Small and rugged
- ☐ Battery powered for up to 8 hours
- ☐ User adjustable sensitivity
- ☐ Standard tripod mounting threads

TIMING PARAMETERS

Sensor Piezoelectric acoustic sensor

INPUT / OUTPUT SIGNALS

Output (BNC) Positive 5V TTL into 50Ω

Pulse width 30mS (Standard)
10mS (Optional)

Trigger indicator LED

Electrical input Battery powered (Built in)
Battery charger included

Low power indicator LED

MECHANICAL

Dimension mm (w/d/h) **Unit**
152mm x 88mm (5.9" x 3.46")
Tripod mount block
16mm high (0.62")

Weight 1Kg (2.2lbs)

Tripod mount 1 x 1/4-20UNC
1 x 3/8-16UNC

ENVIRONMENTAL

Storage temperature -20°C to +50°C

Operating temperature -15°C to +50°C

Charging temperature 0°C to +40°C

Humidity 10—90% RH non condensing

Vibration shock 10—40 Hz Max. 10g in any direction

EMC Meets all UKCA/EU harmonised standards



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Multi-head Intensified Camera System



Up to 8 camera heads per control module

1360 x 1024 pixel, 12-bit sensor resolution

Single or double image capture

5ns minimum shutter

The Specialised Imaging CERBERUS camera system offers framing camera image capture performance with the addition of multiple camera control.

Each camera head can capture either one or two 1.4 MegaPixel 12-bit images with exposure times down to 5ns.

A maximum of 8 Control modules can be connected to allow a total of 64 cameras controlled from a single PC.

The CERBERUS system is flexible enough to allow multiple 3D/Stereoscopic image pairs or sequential images with a 5ns interframe time, equating to 200 Million Frames/second.

FEATURES

- ☐ Control up to 64 camera heads
- ☐ Adjustable exposure down to 5ns
- ☐ Head to Head adjustable interframe time down to 5ns
- ☐ Nikon lens mount fitting
- ☐ Ethernet communications
- ☐ Compact and rugged design

OPTICAL

Lenses	Nikon F-mount (ruggedized mounting system)
Shutter	Electro-mechanical
Distortion	Nominally zero
Intensity variation	Better than 5% across the image

INTENSIFIER / SENSOR

	X-HEAD	D-HEAD
Image Sensor	ICX285AL (Intensified)	
Active CCD Pixel	1360 (H) x 1024 (V) pixels	
Pixel Size	6.45 µm (H) x 6.45 µm (V)	
Dynamic Range	12 bits	
Intensifier	8mm High resolution MCP	
Input window	Fused Silica	
Output window	Fibre Optic	
Photocathode	S25, others on request	
Phosphor screen	P43	P46
Gain	Variable up to 10,000	
Dynamic resolution	50lp/mm	>36lp/mm
Images	Single	Two (550ns interframe time)

MECHANICAL

Dimension mm (w/d/h)	Head (without lens) 9.4cm x 21cm x 9.4cm (3.7" 8.2" x 3.7") Controller 19" rack mount 3U case
Weights	Head 3kg (6.6lbs) Controller 7kg (15.4lbs)
Tripod Mount	3/8-16 UNC Female in head base

ENVIRONMENTAL

Housing	19" Rack Mount 2U case
Storage temperature	-10°C to +50°C
Operating Temperature	-5°C to +40°C
Humidity	10 - 90% RH non condensing
Vibration shock	10 - 40 Hz Max. 10g in any direction
EMC	Meets all UKCA/EU harmonised standards

TIMING PARAMETERS

System Clock	200MHz quartz crystal control
Inherent Delay	500ns
Exposure Mode (each head)	Single exposure or multiple exposures (Max. 8) per head
Exposure Time	5ns - 10ms in 5ns steps
Interframe Time (head-to-head)	5ns - 20ms in 5ns steps
Delay to 1st exposure	500ns - 10ms in 5ns steps
Flash Outputs	5ns to 1ms in 5ns steps
Separation Time	30ns - 20ms in 5ns steps (multiple exposures on same channel)

INPUT / OUTPUT SIGNALS

Trigger 1 (BNC)	Electrical signal Threshold variable from ± 25V Positive or Negative polarity, Make/Break 50Ω or 1KΩ termination
Trigger 2 (BNC)	Electrical signal Threshold variable from ± 25V Positive or Negative polarity, Make/Break 50Ω or 1KΩ termination
Timing Monitor Pulse	Pulse width (min. 5ns) and position user programmable TTL into 50Ω
Flash Trigger Outputs	Pulse width (min. 5ns) and position user programmable TTL into 50Ω
Remote Camera Interface	Data and command transfer via custom 10m cable.
Camera head control	Data and command transfer via 100Mbps Ethernet cable length 10m (standard), other lengths up to 100m available 100FX Fibre optic Ethernet link (up to 2Km) - optional
Software	Custom software compatible with Microsoft Windows Operating Systems for cameracontrol, image data archiving in various file formats.
Electrical input	Mains 100-240V AC 50-60Hz

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FM87429

Laser illumination system



**Up to 400W
lighting power**

**660nm
wavelength**

**Flexible triggering/
synchronisation**

The Specialised Imaging LUX2 laser illumination system provides up to 400W of lighting power at pulse frequencies up to 10MHz or single pulses up to 30μs.

Simple triggering allows the SI-LUX2 to interface with most high-speed cameras ranging from High-speed video to Ultra highspeed framing cameras.

The 2m laser output light guide includes user interchangeable low coherence beam expanders.

FEATURES

- ☐ Low coherence
- ☐ Pulse width from 10ns – 30μs
- ☐ Pulse frequency range from single to 10MHz
- ☐ Compact design



OPTICAL

Beam Expanders	φ25mm and φ50mm versions
Wavelength	660nm ± 20nm
Output power	200W (-10/ + 30%) or 400W (-10/ + 20%)
Power drop	~ 0.2% / μs for pulses less than 5μs

INPUT / OUTPUT SIGNALS

Sync. Input (BNC)	+5V TTL (laser pulse duration = duration of +5V state)
Indicators	Green LED – Laser is powered & ready
Electrical Input	Mains 100–240V AC 50–60Hz

SAFETY FEATURES

Laser rating	Class 3b
Safety features	Key operated master control power on/off Connector for remote Interlock Capping shutter on lens and laser head
Visual indicator	Green LED = Laser is powered & ready Red LED = interlock indicator Laser classification stated on unit

ENVIRONMENTAL

Storage temperature	-10°C to +50°C
Operating temperature	+10°C to +25°C
Humidity	10—90% RH non condensing
Vibration shock	10—40 Hz Max. 10g in any direction
EMC	Meets all UKCA/EU harmonised standards

TIMING PARAMETERS

Max pulse rate	10MHz
Min. single pulse duration	~10ns
Max. single pulse duration	30μs (max. power drop 20%)
Rise time	~10ns (10%...90%)
Fall time	~5ns
Delay	80ns ±10ns between input to start of light pulse (incl. control cable delay)
Jitter	<5ns
Operating limits	Max. 0.03% duty cycle for unlimited operation Max. 100% duty cycle for 30μs laser operation

MECHANICAL

Dimension (w/d/h)	Laser: 6.2cm x 15cm x 3.6cm (6" x 2.4" x 1.4") Master control box: 12cm x 6cm x 12cm (4.7" x 2.4" x 4.7")
Weight	Laser: 0.5Kg (1.1lbs) Master control box: 0.5Kg (1.1lbs)
Light guide	2m
Control cable	2m between control/safety box & laser



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High intensity short duration flash to freeze motion.



The SI-Monoflash spark source has been designed to emit a single high energy spark, typically 6 Joules, with a duration (FWHM) of approximately 300ns. The emitted light has a spectrum from UV to the infrared region and is ideally suited for many high speed photographic applications.

The single spark, at 300ns, will stop the motion of fast subjects and is typically used for shadowgraph photography.

Easy to use, simply point the SI-Monoflash at the object in the camera's field of view and apply a TTL trigger signal.

The SI-Monoflash can be operated in stand-alone mode or several units can be daisy-chained together to provide status information for a multi-station setup.

FEATURES

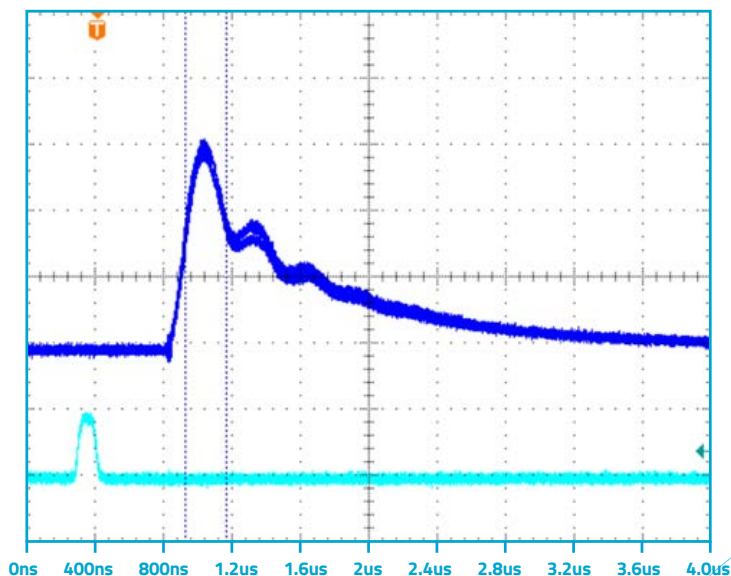
- ☐ 6J in 300ns (typ)
- ☐ Ability to daisy chain multiple units
- ☐ Charge indicator LED indicates when unit is ready to fire.
- ☐ Unit will not trigger until the ready led is lit. This ensures repeatable light output.
- ☐ Charge voltage on flash capacitors is regulated to maintain consistent repeatable light output.
- ☐ Input circuits transient & overvoltage protected.

TRIGGER SOURCE

Input (BNC)	2-5V Positive Edge Triggered
Input Impedance	50 Ω

LIGHT OUTPUT

Duration	Typical 300ns (FWHM)
Capacitance	400nF
Power (max)	6J
Charge Voltage	5KV
Light Source	Spark Gap
Delay (typ)	600ns
Jitter	Less than ± 20 ns
Rise Time (typ)	50ns
Recycle Time	20s approx.



MECHANICAL

Dimension mm (w/d/h)	224 (L) x 125 (W) x 125 (H)
Weight	2.2 Kg/4.85lbs
Housing	Aluminium

ENVIRONMENTAL

Storage temperature	-10°C to +60°C
Operating temperature	-5°C to +45°C
Humidity	10—90% RH non condensing
Vibration shock	10—40 Hz Max. 10g in any direction
EMC	Meets all UKCA/EU harmonised standards

POWER

Main Connector	Switched IEC input socket with built-in RFI/EMI power line filter
	90-250V AC
	50-60Hz
Fuse	2A 20mm (F) Fast blow
Charge Indicator	Illuminated when unit is ready
LED's	Triggering disabled until unit is ready



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Optical Sky Screen Trigger Unit

Rugged Design

Accurate Detection

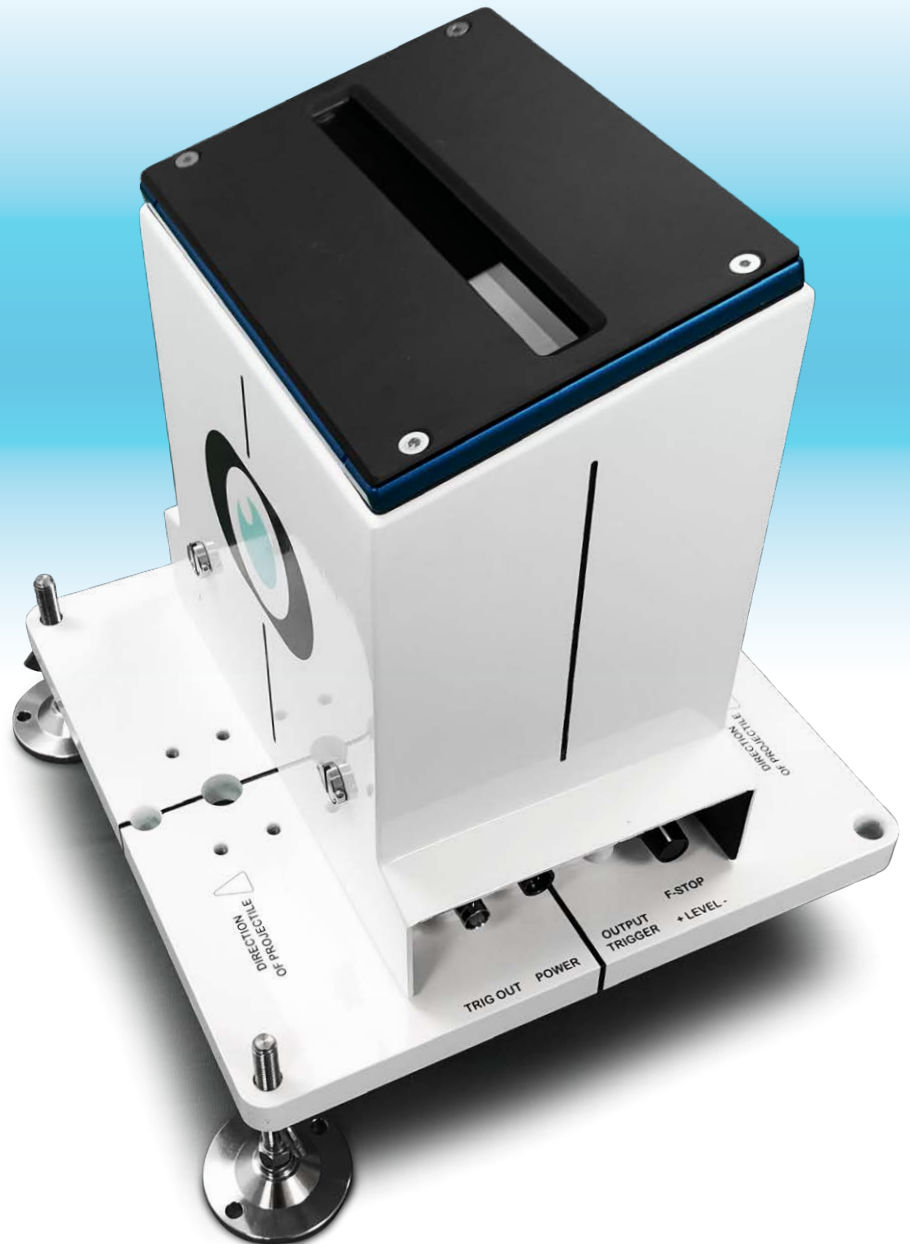
Shadow Detector

The Specialised Imaging SST1 provides a reliable optical trigger for projectile "shadow" detection.

The rugged enclosure allows the SST1 to be used outside in all weather. Our low-ripple AC power unit provides constant power for all day operation.

FEATURES

- ☐ Small and lightweight
- ☐ Low ripple rugged AC power unit
- ☐ Nikon lens mount fitting
- ☐ Local sensitivity adjustment



OPTICAL

Lenses	Nikon F-Mount
Alignment	Engraved Datum lines

SENSOR

Sensor	Multi-Segment Photodiode array. 400nm – 700nm range
--------	--

INPUT / OUTPUT SIGNALS

Output (BNC)	Positive 5V
Trigger indicator	LED
Electrical input	AC Power Unit

POWER

Input	AC Power Unit
-------	---------------

MECHANICAL

Dimension mm (w/d/h)	Unit with Standard cover 28cm x 28cm x 29cm (11" x 11" x 11.5")
	Unit with Extended cover (Optional) 28cm x 28cm x 38cm (11" x 11" x 15")
	AC Power Unit 150mm x 120mm x 80mm 5.9" x 4.7" x 3.1"
Weights	Standard unit 5.8Kg (12.7lbs)
	Extended cover unit (Optional) 6Kg (13.2lbs)
	AC Power Unit 2Kg / 4.4lb

ENVIRONMENTAL

Storage temperature	-10°C to +60°C
Operating temperature	-5°C to +45°C
Humidity	10—90% RH non condensing
Vibration shock	10—40 Hz Max. 10g in any direction
EMC	Meets all UKCA/EU harmonised standards

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Ballistic Velocity Measurement System



3,330 measurements / second

Up to 9,000 measurements storage

5ns Resolution

3 independent input channels

4 independent output channels

The Specialised Imaging VT2 system performs time of flight and/or velocity measurements using 3 input channels.

Capture rates up to 3,330 measurements per second (200,000 rounds per minute) can be measured and up to 9,000 measurements can be stored locally in the Unit head.

The four independent Output channels can be used as a delay generator for other instrumentation to ensure correct triggering of framing, ultra high-speed and high speed video cameras.

FEATURES

- ☐ Velocity measurement
- ☐ Time of flight measurement
- ☐ Ethernet communication
- ☐ Built in 4 channel delay generator
- ☐ Battery powered option
- ☐ 19" Rack mount option

OPTICAL

Measurements	Time of flight or Velocity
Max. rate of capture/fire	200,000 measurements/minute (3,330 Hz)
Max. storage	9,000 measurements (velocity or time of flight)

INPUT / OUTPUT SIGNALS

Input/trigger channels (BNC)	3 (independent) Electrical signal Threshold variable from $\pm 25V$ Positive or Negative polarity, Make/Break 50 Ω or 1K Ω termination
Output channels (BNC)	4 (independent) Pulse width (min. 5ns) and position user programmable TTL into 50 Ω
Control interface	100Mbps Ethernet
Software	Custom software compatible with Microsoft Windows Operating Systems for Control and data archiving.
Electrical input	Mains 100-240V AC 50-60Hz Battery option (inc. charger)

TIMING PARAMETERS

System Clock	200MHz, quartz crystal controlled
Inherent Delay	50ns
Range	5ns to 1s in 5ns steps independently variable
Jitter (trigger)	5ns
Jitter (channel to channel)	<250ps

MECHANICAL

Dimension mm (w/d/h)	360mm x 200mm x 77mm (with handles)
Dimension inches (w/d/h)	14.17" x 7.87" x 3.03" (with handles)
IP rating	IP65
Weight	4Kg (8.8lbs)

ENVIRONMENTAL

Storage temperature	-10°C to +50°C
Operating Temperature	-5°C to +40°C
Humidity	10 - 90% RH non condensing
Vibration shock	10 - 40 Hz Max. 10g in any direction
EMC	Meets all UKCA/EU harmonised standards



Rack mount option

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USA
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Germany
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FM87429

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Supporting you... and your camera

At Specialised Imaging we relish new technological challenges, and we enjoy creating effective solutions. Producing a system that exactly meets your requirements, demands a company prepared and able to create specific optimised solutions.

Specialised Imaging has a strong track record in working with clients to design and develop new functions and facilities that fulfil their requirements.

This level of commitment and support continues throughout the life of your product – on-going advice, problem-solving and the design and reconfiguration of software are all part of our after-sales service.

World-wide partnerships

To provide you with a total ultra-high-speed imaging solution, we have formed strong, strategic relationships with manufacturers in a variety of related fields. These strategic partnerships enable us to offer fully optimised imaging systems that include illumination, optical components, supports and triggering devices.

Close cooperation with our major suppliers also allows us to access and incorporate custom-designed components to specifically enhance the performance of our systems

Our reputation today extends worldwide with customers in North America, Asia-Pacific, Europe and the Middle East regions.

